



UNIVERSITY OF CAPE TOWN

TIME SERIES
PORTFOLIO

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Chapter 1: Foundations

Learning outcomes

1. Explain what a time series is and how it differs from cross-sectional data, and say why temporal dependence calls for its own methods.
2. Treat an observed series as one realisation of a stochastic process.
3. Explain what forecasting is, judge when a process is predictable, and keep forecasts, goals and plans separate.
4. Match a forecast horizon to the decision it supports, and read a forecast as a random variable rather than a single number.
5. Identify trend, seasonality, cycle and irregular variation in a plot, and describe the shapes each of them takes in practice.
6. Produce the four benchmark forecasts, and evaluate any competing model against them on data it has not seen.
7. Use the lag and lead operators, and the algebra built on them, to write models compactly.
8. Define weak stationarity, say which of its conditions a given series violates, and choose a transformation that restores it.
9. Define white noise, MA, AR, ARMA, ARIMA and SARIMA processes, derive their basic properties, and identify them from an ACF and a PACF.
10. Fit, diagnose and forecast these models in R, following the workflow of Box and Jenkins, and test residuals against the white noise standard.
11. Explain volatility clustering and fit ARCH and GARCH models to it.

Lecture 1

Date: 27 July 2026 Topic: Foundations What we did: Questions: What's next?

Example: Continuous series

?@fig-continuous-ts plots the number of monthly airline passengers over a 12 year period. This figure clearly illustrates the concepts of seasonality and trend.

Example: Discrete series

?@fig-discrete-ts plots the changing number of observed seals over a period of 25 years.

IMPORTANT: Time series model dependence through sequential observations.

Assumption: Equally spaced measurements.

Assumption: Autocorrelation across time.

Time series in a single measurement taken at multiple points in time. Technically we are modelling the same thing over and over again (this makes sense why there is sequential correlation).

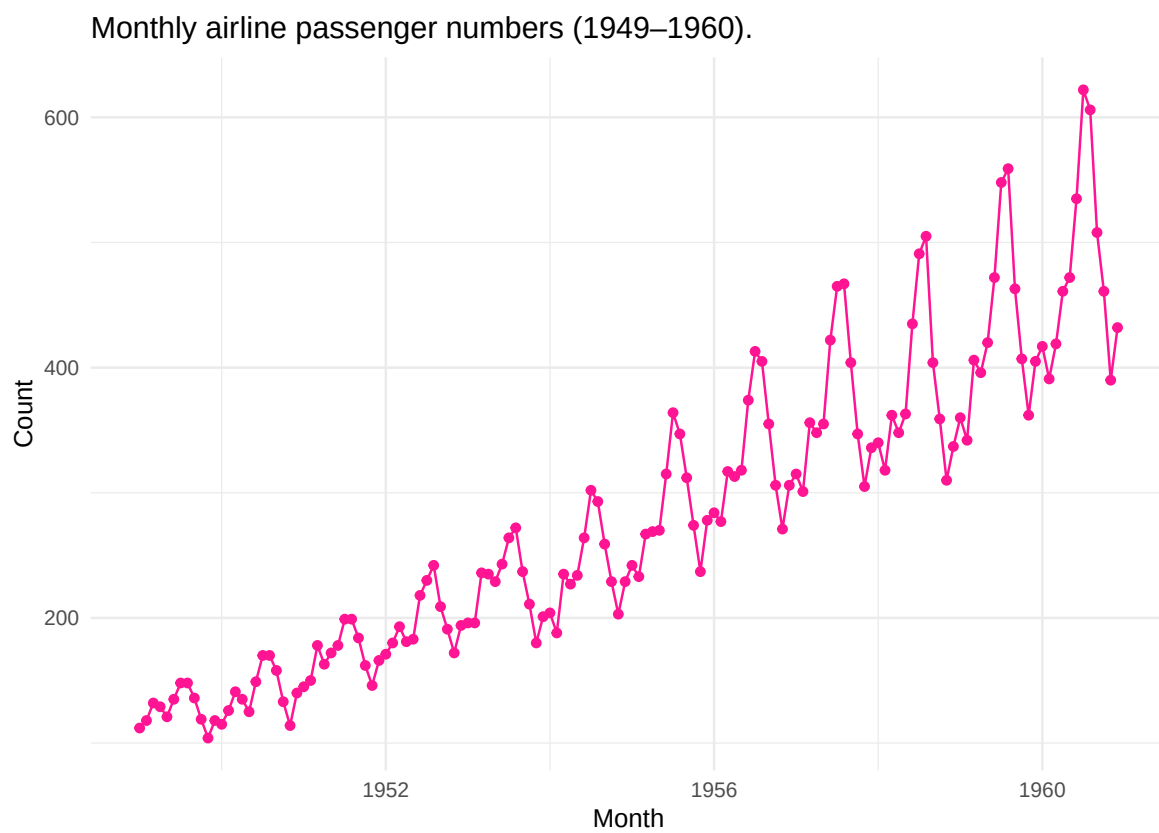


Figure 1: Time series plot of the number of monthly airline passengers from 1949 to 1960.

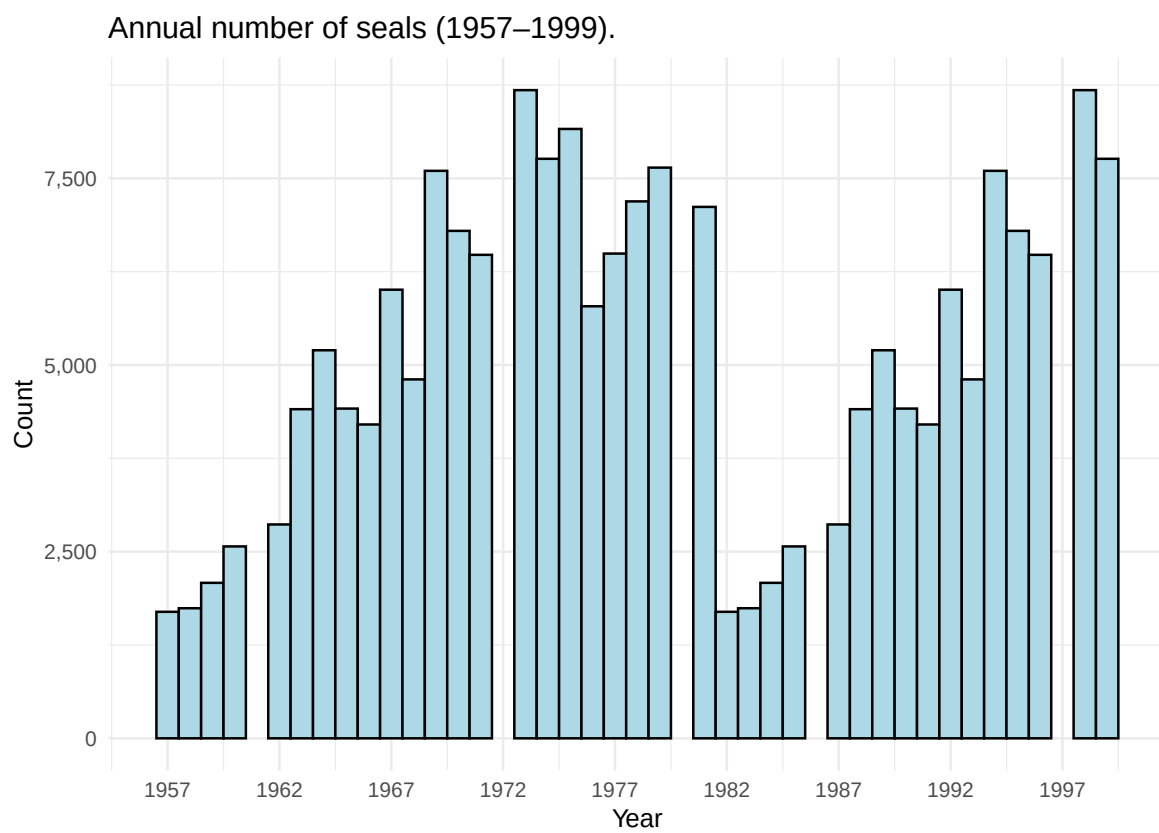


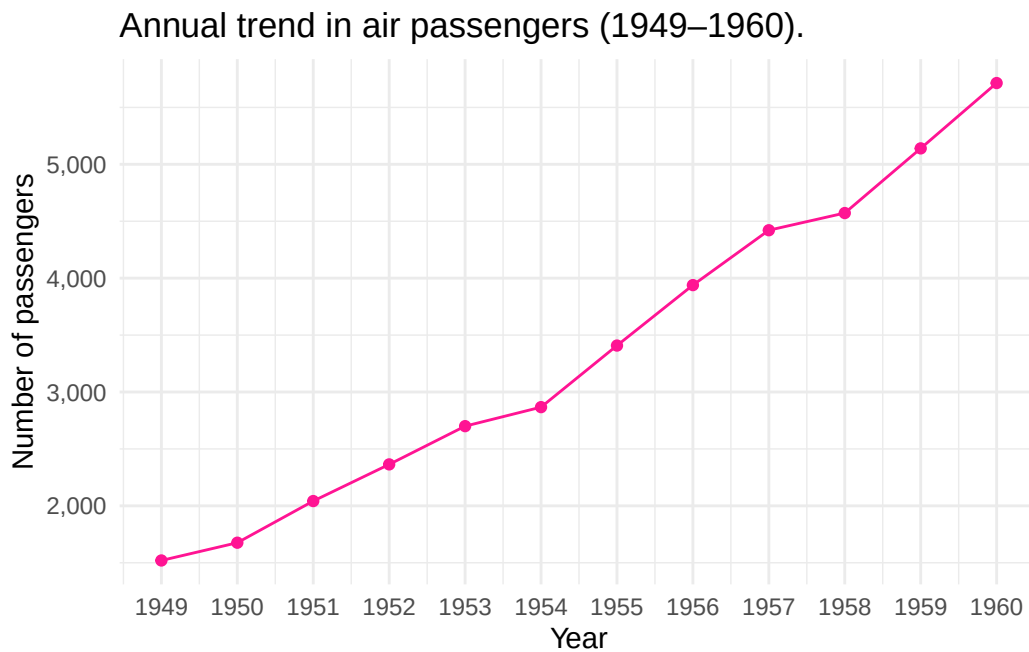
Figure 2: Time series plot of the number of seals from 1975 to 1999.

Lecture 2

Date: 3 August 2026 Topic: What we did: Questions: What's next?

Aggregation

```
1 agg_AirPassengers = aggregate(AirPassengers,
2                               frequency = 1,
3                               FUN = sum)
4
5 aggAP_df = data.frame(
6   year = as.numeric(time(agg_AirPassengers)),
7   count = as.numeric(agg_AirPassengers)
8 )
9
10 ggplot(data = aggAP_df,
11        mapping = aes(x = year,
12                      y = count)) +
13   geom_line(color = 'deeppink') +
14   geom_point(colour = 'deeppink') +
15   scale_x_continuous(breaks = seq(1949, 1960, by = 1)) +
16   scale_y_continuous(labels = scales::comma) +
17   labs(
18     title = "Annual trend in air passengers -(1949-1960).",
19     x = "Year",
20     y = "Number of passengers"
21 ) +
22 theme_minimal()
```



NOTE: Noise in data introduces different realisations. Noise is stochastic.

Chapter 2: The Forecasting Problem

Lecture 3

Date: Topic: What we did: Questions: What's next?

Forecasting: This is kind of like making a prediction. A forecast is a random variable.

Goal setting: This is what we want to happen Planning: This is how we are going to reach our goal

CHANGING THE FORECAST DOESN'T CHANGE WHAT WILL HAPPEN!

A *random walk* is a non-stationary process where memory is permanent. This means that every future shock accumulates endlessly. The next observation is dependent on the previous observation.

A *stationary* process is where memory decays geometrically, shocks die out. This will force the process back to the mean.

Question: What is a mixed model? Answer (19 August 2026): This is a multivariate time series.

Chapter 3: Exploring a Series

Option 1: The time domain

Here we look at previous observations. This leads to AR,MA,ARMA and ARIMA models.

Option 2: The frequency domain

Here we look at previous trends.

seasonality must have a fixed and known interval Question: Otherwise is it a cycle? Answer:

Lecture 4

Date: Topic: What we did: Questions: What's next?

Exercises

1. Why is ordinary random k-fold cross-validation inappropriate for forecasting?

If you exclude an observation at time $t-1$ but use observation t , you are technically predicting the past based on future observations. Future observations must never be used when estimating a forecasting model as this violates a fundamental principal of forecasting -

2. Define temporal leakage.

Future observations enter model estimation. Reported accuracy looks excellent but is unattainable in practice - the future is unknown when real forecasts are made.

3. Contrast Walk_Forward, Expanding Window and Sliding Window validation.

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4. When would Sliding Window outperform Expanding Window?

When data is non-stationary.

5. A firm refits its model daily using only the most recent 250 observations. Which strategy is this?

Sliding window

6. Why must temporal ordering be preserved during validation?

Otherwise the model knows things it could not have known at forecasting time. This leads to inflated accuracy that is unattainable in practise.